DIBYA BHUSAL

Software Engineer · AI/ML · Cybersecurity · Blockchain · Full-Stack · IoT

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EDUCATION

REVA University — B.Tech CSE (IoT, Cybersecurity & Blockchain) Aug 2024 – May 2028

Coursework: Data Structures & Algorithms · Cybersecurity · Blockchain · IoT · Quantum Computing · AI/ML · Software Engineering

- Google Cloud Certified — Prompt Design in Vertex AI (Aug 2025)

EXPERIENCE

Cybersecurity Intern Jun 2025 – Aug 2025

Industrial placement · Bengaluru, India

- Audited live network infrastructure using Nmap and Wireshark; documented 12+ vulnerability findings in a structured pentest report.
- Applied OWASP Top-10 remediation patterns on industrial control system interfaces, receiving a strong written recommendation.
- Built reusable Python scanning scripts that reduced manual audit time by ~40% across two project cycles.

PROJECTS

Consensus Bedrock Engine · AWS Bedrock / Lambda / DynamoDB / React Mar 2025

- Architected a production multi-agent LLM consensus system on AWS: two worker models (Llama 4 Scout 17B, Ministral 8B) scored across a custom 7-metric engine (confidence, hallucination defense, specificity, readability) before a frontier Judge (Amazon Nova Pro) is invoked only on conflict.
- Engineered adaptive throttling, cross-region inference failover, and DynamoDB MD5 semantic cache — achieving <50 ms cached latency and <$0.000006 average cost per query.
- Deployed end-to-end via AWS CDK with Amplify CI/CD; live at prototype.d3ddhsf8bhejkw.amplifyapp.com.

NeighborHealth AI · FastAPI / XGBoost / Supabase / Leaflet.js Apr 2025 | Build for Bengaluru Hackathon

- Built a hyperlocal disease outbreak prediction platform covering 198 BBMP wards x 12 diseases; XGBoost model trained on Karnataka Parliamentary health records (RS Session 255, 2018–2021) achieved ROC-AUC 0.96.
- Engineered a 9-feature vector (rainfall, temp, humidity, historical case counts, community reports, population density) with automated daily GitHub Actions refresh pipeline generating 2,376 risk scores/day.
- Integrated Twilio SMS + Gmail threshold alerts, OpenRouter/Gemini AI per-ward explanations, and a multimodal health checker (skin image triage). Live at neighborhealth-ai.vercel.app.

Phantom (TRI-ZEROTRO) · LangGraph / Solana / FastAPI / IPFS / pgvector May 2025 | Hackathon

- Co-built a 3-platform autonomous trust ecosystem: PHANTOMID (AI identity defense), TRUSTFLOW (blockchain escrow + AI work verification), and WITNESSCHAIN (deepfake-resistant civic evidence network).
- Designed LangGraph AI agents that cross-reference 15B+ leaked credentials via HIBP, NVD CVE, and CERT-In feeds — anchoring every threat event immutably on Solana via Anchor smart contracts.
- Integrated pgvector semantic search, IPFS decentralized storage, and TensorFlow.js on-device deepfake detection; live prototype at phantom-sigma-rust.vercel.app.

Aptos Synapse · Aptos Blockchain / Node.js / Ollama / LLM Agents 2025 | Aptos Hackathon

- Designed a decentralized AI Agent Delegation Protocol on Aptos — six specialized local LLMs (LLaMA, Mistral, Qwen) operate as independently verifiable agents, each performing work, getting evaluated, and earning rewards autonomously.
- Architected the full agent orchestration layer, Aptos smart contract payment release logic, and high-performance frontend; system demonstrates a functioning AI labor marketplace on-chain.

TECHNICAL SKILLS

Languages:	Python · JavaScript (ES2022) · C · C++ · Solidity · Rust (basic) · SQL · Bash
AI / ML:	XGBoost · Scikit-learn · LangGraph · Groq · Ollama · OpenRouter · Vertex AI · pgvector · TensorFlow.js
Web / Backend:	FastAPI · Node.js · Express · React · Next.js · Leaflet.js · REST APIs · Pydantic v2 · WebSockets

Cloud / Infra:	AWS (Bedrock, Lambda, CDK, Amplify, DynamoDB) · Supabase · Vercel · IPFS · Docker · GitHub Actions
Blockchain:	Solana · Anchor · Aptos · Web3.js · Solidity · Smart Contracts · IPFS · Hardhat
Cybersecurity:	Nmap · Wireshark · Kali Linux · OWASP Top-10 · AES-256 · SHA-256 · Penetration Testing · HIBP · NVD CVE
IoT / Embedded:	ESP32 · Arduino IDE · MQTT · C++ firmware · Capacitive Sensing · Relay Circuits
Tools:	Git · GitHub · VS Code · Linux · Postman · Firebase · Supabase Realtime

ACHIEVEMENTS & LEADERSHIP

- Google Cloud Certified — Prompt Design in Vertex AI (2025).
- Consistently ranked among top-performing teams across 15+ national and university hackathons (AI, cybersecurity, blockchain, full-stack), including multiple top-5 finishes in engineering and innovation tracks.
- State-Level Cricket Player (U16 & U19, Nepal) — represented state in national-level tournaments.
- Led 30-member university tech/coding club; ran weekly sessions on cybersecurity, systems programming, and applied AI.